

Surya Vara Prasad Alla

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SUMMARY

Computer Vision Engineer with 4+ years of combined experience in software engineering and deep-learning-based perception systems. I specialize in designing and deploying high-impact vision pipelines for detection, 3D geometry, OCR, and edge/GPU environments, with a strong product mindset and a track record of measurable gains in accuracy, reliability, and system performance.

SKILLS

Programming: Python, C++

CV & DL: PyTorch, TensorFlow, YOLOv5-v9, OpenCV, Open3D, NumPy, Point Cloud Processing

Deployment: CUDA, TensorRT, Jetson, Docker, FastAPI, AWS SageMaker

Data: Image/Video Preprocessing, 3D Reconstruction, OCR

EXPERIENCE

Fraunhofer FIT

AI Software Engineer

Hamm, Germany

Jan 2024 – Present

- Delivered production-grade CV pipelines, including a DINOv2 + CountGD poultry-counting system achieving **95% accuracy** and reducing manual inspection time by **70%**.
- Engineered LCNN + SAM based foot-geometry extraction with custom preprocessing and post-processing, significantly improving keypoint precision and stability.
- Fine-tuned YOLOv9-e for microscopy-based cancer cell detection, achieving **90%+ accuracy** in clinical workflows.
- Developed a tile-based OCR pipeline for A2 engineering drawings, reducing manual correction workload by **30%**.
- Optimized and deployed CV models to edge/GPU hardware (Jetson, CUDA, TensorRT), improving inference latency and enabling real-time applications.
- Built reproducible ML workflows using **DVC**, **MLflow**, **GitLab CI**, **Docker** for traceable experimentation and stable deployment pipelines.
- Designed automated evaluation utilities to benchmark accuracy, latency, and robustness across CV model variants.

InnoSaddle GbR

ML Engineer & Technical Co-founder

Siegen, Germany

Jul 2022 – Dec 2023

- Processed 3D LiDAR/TrueDepth scans using Open3D for biomechanical back-geometry analysis to support a data-driven saddle-fit product.
- Designed 3D reconstruction workflows combining geometric processing and photogrammetry-inspired techniques to improve surface accuracy and measurement consistency.
- Led technical development and collaborated with biomechanics experts to translate CV insights into product features and validation workflows.

Wipro Technologies

Software Engineer (Data)

Hyderabad, India

Feb 2018 – Aug 2019

- Designed data pipelines to transform multi-device Teams JSON logs into structured analytics datasets in Azure Data Lake.
- Built Azure Data Factory ETL workflows integrating USQL, Databricks (Scala), and SQL Server for reliable daily processing.
- Improved ingestion throughput and system stability through pipeline optimization and operational monitoring.

EDUCATION

University of Siegen

M.Sc. Mechatronics | Specialization: Computer Vision, Deep Learning, C++

Siegen, Germany

Oct 2019 – Sep 2023

Osmania University

B.E. Mechanical Engineering

Hyderabad, India

Sep 2013 – Mar 2017

AWARDS & ACHIEVEMENTS

Awarded the **Gründungsstipendium NRW** (€28,800, 2022) by Projektträger Jülich for developing an innovative AI/computer-vision-based startup concept.